

Eastern Visayas State University

School of Engineering

CAPSTONE PROJECT & RESEARCH I

Proposed Capstone Project:

LegiSys: Municipal Legislative Management System of Tolosa

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Technology Comparison Table

Criterion \ Technology	Technology A: Electronic Document Management and Tracking System for PhilHealth Regional Office NCR	Technology B: LemTrac: Legislative management and Tracking System	Technology C: Web-Based Document Tracking System with QR-Code Technology	Gap Identified / Opportunity for Capstone Project
Primary Purpose/Problem Solved	Manages and transmits various documents. Centralizes storing information and electronic documents through a web application to quickly retrieve, share, disseminate, and monitor the status of documents.	Developed to convert manual municipal legislation processes into a digital format, solving the critical problems of delayed document processing time and misplaced files that could not be tracked by staff.	Developed to improve document tracking, reduce misplaced files, and speed up document retrieval through QR-code scanning	Gap Identified While current systems successfully offer centralized web storage, domain-specific legislative tracking, or QR-code retrieval, they remain fragmented; they fail to combine automated physical-to-digital tracking with a dynamic, user-configurable approval workflow and intelligent data extraction. Opportunity for the Capstone Project This gap presents an opportunity to develop a unified, web-based document management system that integrates dynamic workflow routing and mobile QR-code scanning to automate physical status updates during handoffs, while leveraging smart data extraction to eliminate manual metadata entry.
Core Features/Functionality	Web-based document creation, routing, publishing, and status monitoring. User management, location library management, document type management, action	LeMTrac features eight key integrated modules including session, file, user, and attendance management, alongside a central dashboard, payment processing via POPS, a Local ID System	Document registration, QR-code generation, document status tracking, report generation, user management, and document monitoring.	Gap Identified Although existing systems provide robust individual features such as full digital document creation and publishing, specialized SMS trackers with payment integration, and mobile QR-code scanning they

	management, and document management modules. Centralized electronic storage for faster retrieval, sharing, dissemination, and monitoring.	(LIDS), and an SMS-enabled legislative tracker.		operate in isolation; they lack a unified ecosystem that can seamlessly bridge mobile hardware scanning, complex multi-module administrative tracking, and automated document generation into a single, adaptable platform. Opportunity for the Capstone Project This provides an opportunity to develop an all-in-one Document Lifecycle and Legislative Management System that unifies digital creation and publishing with mobile QR-code scanning for real-time status updates, while integrating automated SMS notifications and modular administrative libraries to fully digitalize both physical handoffs and office workflows.
Target User/Audience	The main users are staff at the PhilHealth Regional Office of NCR, especially those who create, route, process, and manage internal documents such as advisories and corporate office orders.	The target users are the Sangguniang Bayan (SB) members, secretaries, and office staff within the local government units of Nabua and Bula in Camarines Sur, Philippines.	University clerks, administrative staff, and system administrators.	Current systems are built either for general clerks (Technology C) or corporate staff (Technology A). There is a gap in tools designed to serve both high-level executive decision-makers (who need budget dashboards) and administrative staff (who handle daily document routing) simultaneously.
Technology Stack/Approach	The system was developed using PHP Laravel for the framework, MySQL for storage, and Apache as the web application server.	Using a modified Princeton Project Management Methodology, the web application was built with PHP, CSS, Bootstrap, a MariaDB database, and an Apache web server deployed over a local area network.	HTML, CSS, JavaScript, PHP, Apache Server, centralized database, and QR-code technology.	The compared technologies rely on standard web frameworks (PHP, Laravel, MariaDB/MySQL) and local server stacks (Apache, XAMPP), leaving an unexploited opportunity to introduce automated text-extraction tools (like OCR libraries) to improve data entry efficiency.
Key Strengths	Solves missing documents, poor tracking, ineffective dissemination, and retrieval problems. Centralizes documents and improves productivity and performance in document handling Has clear operational modules that support office workflow.	The system achieved exceptionally high acceptability ratings from users and IT experts due to its strong usability, cross-browser compatibility, and built-in error detection capabilities.	Real-time tracking, reduced human error, improved transparency, efficient retrieval, and compliance with RA 11032.	Current strengths are split: some systems are highly accepted for their workflows (Technology B) and others for their speed (Technology C). The gap lies in creating a unified system that gains high acceptability by offering both high usability and automated physical tracking.

Key Limitations/Weaknesses	The available abstract does not list detailed weaknesses, usability issues, or failed features. No mobile app, QR/OCR, or legislative-specific functions are mentioned in the public summary. The scope appears limited to PhilHealth PRO NCR internal document workflows.	LeMTrac is highly dependent on continuous integration with external modules, specifically requiring data from the Local ID System and Point of Payment System to execute its primary functions.	Limited to document monitoring and tracking processes. No legislative document categorization, session management, or financial monitoring.	Technology B is bottlenecked by its heavy dependence on external systems (LIDS/POPS), while Technologies A and C are limited by their narrow, generic scopes. This highlights an opportunity for a self-contained yet multi-functional system architecture.
Data Sources/Inputs	Includes advisories, corporate office orders, routing information, document type data, action/status entries, user information, and location records.	System inputs consist of LIDS user verification accounts, POPS transaction details, member attendance logs, and legislative document data such as ordinance statuses and author details.	Document information, tracking logs, QR-code data, document status updates, and user records.	Current systems treat tracking logs, attendance data, and financial records as separate data streams. There is an opportunity to process a multi-dimensional data schema where a single document input automatically updates tracking, session, and financial data stores.
Performance Metrics (if quantifiable)	It does not provide numeric usability, efficiency, or speed metrics	The software achieved a "Highly Acceptable" overall ISO 9126 quality metric score of 4.72 on a Likert scale, highlighted by a peak usability score of 4.86 from end-users.	Achieved an overall ISO/IEC 25010 evaluation mean score of 4.94 (Excellent).	While systems boast high ISO usability scores, there is a gap in establishing clear, quantifiable metrics that measure how much time a system saves in meeting specific legal compliance frameworks, such as RA 11032.
Cost/Accessibility	No cost information was seen in the article, since it is web-based	The project was fully funded by the DOST-PCIEERD and designed to provide affordable, local network-based document accessibility to overcome the high costs traditionally associated with ICT adoption in municipalities.	Web-based and accessible through browsers, reducing dependency on manual logbooks.	High-quality tracking often requires expensive ICT infrastructure or cloud deployment. There is an opportunity to design a deeply integrated, feature-rich legislative and financial system optimized specifically for low-cost, local-network deployments in smaller municipalities.
Scalability	It suggests that it can support more documents and routing activities than manual filing systems. However, the multi-office scaling, cloud deployment, or load testing was not described in the abstract	While initially soft-launched in two specific towns, the system's database complies with national DICT data standards to ensure smooth future modular expansion and replication across other municipalities.	Can manage large volumes of documents through a centralized database and server architecture.	Many existing systems lack detailed cloud-ready or multi-office scaling strategies in their design. A gap exists for systems built from the ground up to comply with national data standards, ensuring easy future expansion into broader modular networks.
Security/Privacy Concerns	The system includes user management and is intended to help securely store, route, and retrieve internal documents	To protect against digital vulnerabilities and malware, the system enforces strict role-based access control where standard users have view-only privileges while administrators manage the secure server database hooks.	Uses user authentication and controlled access to documents. Security was rated "Excellent" in evaluation.	While basic user authentication exists, standard tracking systems do not address the complex privacy needs of separating sensitive legislative drafts and internal budget details through distinct, fine-grained role permissions.
Environmental/Social Impact	Mainly organizational, reducing paper handling, workflow efficiency, and faster retrieval of records.	The system promotes good governance and reduces bureaucracy while generating	Reduces paper-based tracking, improves transparency, accountability, and	systems focus purely on internal paper reduction and office productivity. There is an unaddressed

		positive resource savings by shifting toward paperless operations that stop the costly cycle of replacing misplaced documents.	public service efficiency.	opportunity to design a tracking system that directly uses its audit trails to promote institutional transparency and reduce bureaucratic delays for the public.
Future Work/Recommendations	The available summary does not explicitly list future work or recommendations.	The authors explicitly recommend replicating and implementing the LeMTrac system across other municipal local government units alongside continuous training and strategic planning to maintain administrative success.	Recommended mobile access, notifications, external agency integration, and enhanced document management capabilities.	A consistent limitation and future recommendation across existing systems is the lack of real-time mobile integration and automated cross-agency communication, representing a significant opportunity for future development.

References

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